Day 4 — Both Numbers Above 100

Module: Nikhilam Multiplication · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will solve any 2-digit × 2-digit (or 3-digit) multiplication where both factors are slightly *above* 100 (101–112) using the Nikhilam method.

Materials

- Whiteboard with the vertical-line template.
- Day 3 formative quiz returned with marks.
- Drill sheet (10 problems with both factors above 100).
- Stopwatch per pair.

Lesson flow

Warm-up (5 min)

- **Complement-to-100 round** (same as Day 3).
- Add the new round: **excess from 100.** Teacher says “103,” class says “+3.” “108” → “+8.” “112” → “+12.” “101” → “+1.”
- Hand back yesterday’s formative.

Core (15 min) — Both above 100: use excess, cross-ADD

1. **The key change:** instead of *deficit* (negative), we now have *excess* (positive). Instead of cross-subtract, we **cross-add**.

2. **The template** (vertical line, base = 100):

```
  103 │ +3       ← excess from 100
× 104 │ +4
──────┼─────
   ?  │  ?
```

3. **Walk through `103 × 104`:**

   - Excesses: +3, +4.
   - LEFT: 103 + 4 = **107**. (Or 104 + 3 = 107. ✓ Either cross.)
   - RIGHT: 3 × 4 = **12**. Pad to 2 digits: `12`.
   - Answer: **10712**. ✓ (Calculator check: 103 × 104 = 10712.)

4. **Walk through `108 × 107`:**

   - Excesses: +8, +7.
   - LEFT: 108 + 7 = **115**.
   - RIGHT: 8 × 7 = **56**. Pad to 2 digits: `56`.

    - Answer: **11556**. ✓
5. **The pattern:**

Both below	Both above
Cross-**subtract**	Cross-**add**
Deficit × deficit	Excess × excess
Right part is product of POSITIVE numbers (deficits)	Right part is product of POSITIVE numbers (excesses)

In both cases the right part is a positive product. The sign issue only appears in the *mixed* case (Day 5).

6. **The overflow case (rare here): `112 × 113`.**
    - Excesses: +12, +13.
    - LEFT: 112 + 13 = **125**.
    - RIGHT: 12 × 13 = **156**. Right part should be 2 digits; the leading 1 overflows.
    - **Carry:** left = 125 + 1 = **126**; right = `56`.
    - Answer: **12656**. ✓

Activity (20 min) — Drill

Drill sheet (10 problems):

```
1.  103 × 104        2.  102 × 105        3.  108 × 107
4.  109 × 102        5.  101 × 106        6.  111 × 104
7.  112 × 113        8.  106 × 109        9.  103 × 103
10. 105 × 105
```

- 5 min individual silent.
- 5 min paired check.
- 10 min mixed: alternate problems from yesterday (both below) and today (both above) on the same drill sheet — call it the "mixed-base" sheet. Students need to first **decide** whether each is "both below" or "both above" before applying the right cross-operation.

Answer key (teacher): | Q | Sign | Cross | Left | Right | Answer | |—|——|——-|——|——-|——–| | 1 | both above | + | 107 | 12 | 10712 | | 2 | both above | + | 107 | 10 | 10710 | | 3 | both above | + | 115 | 56 | 11556 | | 4 | both above | + | 111 | 18 | 11118 | | 5 | both above | + | 107 | 06 | 10706 | | 6 | both above | + | 115 | 44 | 11544 | | 7 | both above | + | 125+1=126 | 56 | 12656 | | 8 | both above | + | 115 | 54 | 11554 | | 9 | both above | + | 106 | 09 | 10609 | | 10 | both above | + | 110 | 25 | 11025 |

Wrap-up (5 min)

- **Formative quiz Part B (3 min)** — see `quizzes/formative.md`.
- Preview Day 5: *"Tomorrow — what happens when ONE number is above and ONE is below? Sign care needed."*

Student-facing content

Both numbers above 100

Step	What you do
1	Find each number's **excess** above 100 (positive number).
2	**Cross-ADD** for the LEFT part.
3	**Multiply** the two excesses for the RIGHT part.
4	**Pad the RIGHT part to 2 digits.**
5	If the RIGHT part overflows, **carry** the overflow into the LEFT.

Examples: - 103 × 104 → 107 | 12 → **10712** - 108 × 107 → 115 | 56 → **11556** - 112 × 113 → 126 | 56 → **12656** (right part overflowed)

Same sūtra, mirror operation: when deficits become excesses, subtraction becomes addition.

Homework

- 10 mixed problems (both-below AND both-above mixed). Decide first, then solve. Bring tomorrow.

Differentiation

- **Tier up:** Try `1003 × 1004` (base 1000). The right part should be 3 digits. Confirm `1003 × 1004 = 1007012`.
- **Tier down:** Stay with `103 × 104`, `102 × 103`, `104 × 105` until comfortable.

Teacher notes

- The "decide which sign first" step on the mixed drill sheet is the most important habit-builder of this day. Reinforce: *before* you start cross-adding or cross-subtracting, **glance at both numbers** and decide whether they're both below, both above, or split.
- A few students will try to carry into the right part from the left (instead of the other way). Walk through the place-value reasoning: a "100" overflow on the right is *one more in the next column up*, which is the left part.

Citation(s) used in this lesson

- *Vedic Mathematics Batch1.pdf* (Vedic Cultural Center, Sammamish WA) — Nikhilam multiplication, "above the base" examples.
- *Vedic_Maths.pdf* (ISKCON Desire Tree) — practice problems for above-base multiplication.